## Mean, Variance, Standard deviation and Box plot

Q1. To compute the **mean (average)** of the given age data:

**Step 1: Add all the values together**

24+25+26+23+24+22+27+24=195

**Step 2: Divide by the total number of values**

There are **8** data points.

$$\text{Mean} = \frac{195}{8} = 24.375$$

✅ **Final Answer:**

**Mean = 24.375**

To compute the **variance** of the data:
**Data:** 24, 25, 26, 23, 24, 22, 27, 24
**Mean (from before):** 24.375

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**Step 1: Find the squared difference from the mean for each value:**

$$(24 - 24.375)^2 = (-0.375)^2 = 0.1406$$
$$(25 - 24.375)^2 = (0.625)^2 = 0.3906$$
$$(26 - 24.375)^2 = (1.625)^2 = 2.6406$$
$$(23 - 24.375)^2 = (-1.375)^2 = 1.8906$$
$$(24 - 24.375)^2 = (-0.375)^2 = 0.1406$$
$$(22 - 24.375)^2 = (-2.375)^2 = 5.6406$$
$$(27 - 24.375)^2 = (2.625)^2 = 6.8906$$
$$(24 - 24.375)^2 = (-0.375)^2 = 0.1406$$

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**Step 2: Sum of squared differences:**

0.1406+0.3906+2.6406+1.8906+0.1406+5.6406+6.8906+0.1406=17.8748

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**Step 3: Divide by n or n−1**

- For **population variance**, divide by **n = 8**

$$\text{Population Variance} = \frac{17.8748}{8} = 2.23435$$

- For **sample variance**, divide by **n-1 = 7**

$$\text{Sample Variance} = \frac{17.8748}{7} = 2.5535$$

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✅ **Final Answer:**

- **Population variance = 2.23** (rounded to 2 decimal places)
- **Sample variance = 2.55** (rounded to 2 decimal places)

Q2. Let's compute the **mean** of the following data:

**Data:**

24, 25, 26, 23, 24, 22, 27, 24, 60

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**Step 1: Add all values together**

24+25+26+23+24+22+27+24+60=255

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**Step 2: Count the number of values**

There are **9** data points.

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**Step 3: Compute the mean**

$$\text{Mean} = \frac{255}{9} \approx 28.33$$

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✅ **Final Answer:**

**Mean = 28.33** (rounded to 2 decimal places)

Let's compute the **variance** for the data:

**Data:**

24, 25, 26, 23, 24, 22, 27, 24, 60
**Mean = 28.33** (from previous calculation)

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**Step 1: Compute squared differences from the mean**

$$(24 - 28.33)^2 = (-4.33)^2 = 18.7489$$
$$(25 - 28.33)^2 = (-3.33)^2 = 11.0889$$
$$(26 - 28.33)^2 = (-2.33)^2 = 5.4289$$
$$(23 - 28.33)^2 = (-5.33)^2 = 28.4089$$
$$(24 - 28.33)^2 = (-4.33)^2 = 18.7489$$
$$(22 - 28.33)^2 = (-6.33)^2 = 40.0689$$
$$(27 - 28.33)^2 = (-1.33)^2 = 1.7689$$
$$(24 - 28.33)^2 = (-4.33)^2 = 18.7489$$
$$(60 - 28.33)^2 = (31.67)^2 = 1002.5889$$

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**Step 2: Sum of squared differences**

$$\text{Sum} = 18.7489 + 11.0889 + 5.4289 + 28.4089 + 18.7489 + 40.0689 + 1.7689 + 18.7489 + 1002.5889 = 1145.6001$$

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**Step 3: Divide by n (population) or n - 1 (sample)**

- **Population variance**:

$$\frac{1145.6001}{9} \approx 127.29$$

- **Sample variance**:

$$\frac{1145.6001}{8} \approx 143.20$$

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✅ **Final Answer:**

- **Population variance ≈ 127.29**
- **Sample variance ≈ 143.20**

Let's interpret the **variance** and **standard deviation** using your age data:

**Data:**

24, 25, 26, 23, 24, 22, 27, 24, 60
**Mean = 28.33**
**Sample variance ≈ 143.20**
**Sample standard deviation = √143.20 ≈ 11.96**

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### 🔍 What does variance mean in this context?

- **Variance** measures how spread out the ages are from the mean (average).
- A **higher variance** means the values are **more spread out**.
- In this case, the variance is **143.20**, which is relatively high **because of the outlier: 60**.
- Most of the ages are in the 22–27 range, but 60 is far from the mean and increases the variance significantly.

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### 📏 What does standard deviation mean?

- The **standard deviation** is the square root of the variance and gives a measure of spread **in the same unit** as the data (here, **years**).
- A standard deviation of ≈ **11.96 years** means that, on average, the ages deviate about 12 years from the mean of 28.33.
- Again, this large value mostly results from the single outlier (60).

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### 📌 Interpretation Summary:

- Most of the ages are around 24–27 years, but the **presence of the outlier (60)** causes the **mean to increase** and the **variance and standard deviation to be large**.
- If that outlier were removed, you'd likely see a **much lower variance and standard deviation**, indicating that the ages are more closely grouped around the mean.

Q. Let's **draw a box plot** for the data:
**Data:** 24, 25, 26, 23, 24, 22, 27, 24, 60

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**Step 1: Sort the data**

Sorted data:
**22, 23, 24, 24, 24, 25, 26, 27, 60**

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**Step 2: Find the five-number summary**

1. **Minimum:** 22
2. **Q1 (1st Quartile):** Median of lower half = median of (22, 23, 24, 24)

$$Q1 = \frac{23 + 24}{2} = 23.5$$

3. **Median (Q2):** Middle value = **24**
4. **Q3 (3rd Quartile):** Median of upper half = median of (25, 26, 27, 60)

$$Q3 = \frac{26 + 27}{2} = 26.5$$

5. **Maximum:** 60

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**Step 3: Calculate the Interquartile Range (IQR)**

IQR=Q3−Q1 = 26.5 - 23.5 = 3

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**Step 4: Determine outlier boundaries**

- **Lower Fence/bound** = Q1 – 1.5 × IQR = 23.5 – 4.5 = **19**
- **Upper Fence/bound** = Q3 + 1.5 × IQR = 26.5 + 4.5 = **31**

Any data **above 31 or below 19** is considered an outlier.
✅ So, **60 is an outlier**.

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**Step 5: Draw the box plot**

I'll represent it **textually** here:

Outlier: *

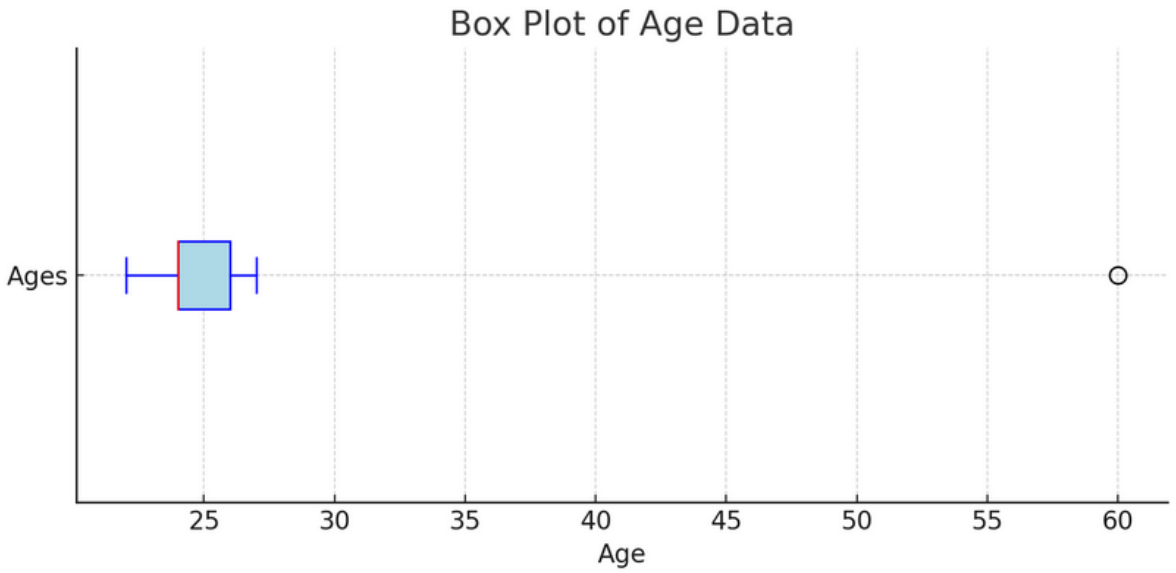
```
        |
        |
22  23.5  24    26.5    27    60
 |----|=====|=======|-----|    *
    Q1   Median  Q3
```

- Box spans from **Q1 (23.5)** to **Q3 (26.5)**
- **Median = 24** is inside the box
- **Whiskers** extend from **Min = 22** to **Max before outlier = 27**
- **60** is shown as an **outlier**

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Here is the box plot for your age data. You can clearly see:

- The **box** spans from Q1 (23.5) to Q3 (26.5),
- The **red line** shows the median (24),
- The **whiskers** extend to the minimum (22) and to the last non-outlier (27),
- The **dot at 60** is an **outlier**.




Let's distinguish **empirical formulas** from **other (theoretical or exact) mathematical formulas**:

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## 🔍 Empirical Formula vs Theoretical Formula

| Feature | Empirical Formula | Theoretical/Mathematical Formula |
|---|---|---|
| **Definition** | Based on **observed data or experiments** | Based on **mathematical derivation or laws** |
| **Accuracy** | **Approximate**, works well in practice | **Exact**, under defined conditions |
| **Origin** | Discovered by **fitting data** (data-driven) | Derived using **logic, math, or physical laws** |
| **Example (Statistics)** | Empirical Rule: 68-95-99.7 rule for normal distribution | Standard deviation formula: $\sqrt{\frac{1}{n}\sum(x_i - \bar{x})^2}$ |
| **Example (Physics)** | Newton's cooling law (from experiment) | F = ma (from Newton's 2nd law — theoretical) |
| **When used** | When theory is too complex or unknown | When underlying theory is well understood |
| **Flexibility** | Can change with new data | Fixed — valid as long as assumptions hold |

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## ✅ In Simple Terms:

- **Empirical formulas**:
  → **"We observed this pattern in the data"**
  → Example: "Most people's heights are within 2 SDs of the mean"
- **Theoretical formulas**:
  → **"We proved this using laws or equations"**
  → Example: Area of circle = $\pi r^2$ — no data needed, it's proven

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## 📌 When do we use empirical formulas?

- When it's hard or impossible to derive a formula from first principles.
- When you want a **quick approximation** (e.g., engineering, biology, machine learning).
- When working with **real-world messy data** (like weather forecasting).

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